

DETAILED DESCRIPTION OF THE INVENTION

Defensive Broadening and Future-Proofing Provisions

The embodiments described herein are illustrative and not limiting. The inventive concepts encompass any system, device, method, or computational architecture capable of acquiring radial distance measurements relative to penile anatomy and correlating such measurements with positional indexing data sufficient to reconstruct a structured anatomical geometry profile.

While time-of-flight (ToF) ranging and optical motion sensing are described in various embodiments, alternative depth-sensing technologies, displacement detection systems, hybrid sensing modalities, or future-developed spatial measurement techniques capable of achieving equivalent synchronized distance and positional acquisition fall within the scope of the present disclosure.

Similarly, reconstruction methodologies may employ deterministic mathematical modeling, statistical interpolation, adaptive filtering, machine learning regression, neural surface estimation, physics-based tissue modeling, or future-developed computational paradigms without departing from the underlying concept of generating a user-specific penile anatomical geometry profile from synchronized spatial sampling.

The geometry profile output may be represented in raw, normalized, encoded, abstracted, encrypted, tokenized, hashed, vectorized, or otherwise transformed formats. The disclosure encompasses representations suitable for direct visualization, secure storage, compatibility scoring, device calibration, research analytics, or integration within distributed biometric ecosystems.

Sensor counts, sensor positioning, sampling density, signal wavelength, communication protocol, processing location, encryption method, calibration approach, housing geometry, material composition, or regulatory configuration may vary across implementations. Any embodiment preserving the functional capability of generating a structurally indexed penile anatomical geometry profile using coordinated spatial measurement techniques remains within the scope of the invention.

Future hardware advancements, including miniaturized depth arrays, solid-state lidar systems, embedded photonic sensors, advanced motion tracking systems, or integrated bio-signal fusion modules, may be incorporated without altering the foundational inventive architecture disclosed herein.

The present disclosure therefore contemplates both currently available sensing and computational technologies and future-developed systems capable of performing synchronized radial distance acquisition and longitudinal positional correlation to generate user-specific penile anatomical geometry profiles.